GigaWhat Energy

GigaWhat Electricity
Controlled procedure

PR-ELEC-015 Switching Programmes on the 11 kV and 33 kV Networks

Field	Value
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Owner	Head of Electricity Network Operations
Approved by	Electricity Safety Manager
Applies to	All GigaWhat Electricity sites: 11 kV and 33 kV switchgear, feeders, primary transformers and grid transformers

FICTIONAL DOCUMENT. Created for the GigaWhat demo. Not for real operational use.

1 Purpose

This procedure sets out how switching programmes for the 11 kV and 33 kV networks are written, checked, approved and carried out, so that every switching operation is planned, confirmed with the Control Engineer and recorded.

2 Scope

This procedure applies to planned and fault switching on 11 kV and 33 kV switchgear, feeders, primary transformers and grid transformers at all GigaWhat Electricity sites. This includes SWG-RVL-33 and the 33 kV connections of GT-1 and GT-2 at ESS-RVL, SWG-ASH-11 at ESS-ASH and SWG-KGM-11 at ESS-KGM. Switching on the 132 kV side of GT-1 and GT-2 is outside this procedure.

3 References

- PR-ELEC-001 Distribution Safety Rules - Summary for Operational Staff
- PR-ELEC-012 Isolation and Earthing of 11 kV Circuits
- PR-ELEC-020 Permit-to-Work and Sanction-for-Test (Electrical)
- PR-ELEC-047 Response to Transformer Temperature Alarms
- PR-ELEC-075 Fault Restoration on 11 kV Feeders
- PR-CORP-001 Reporting Incidents and Near Misses
- HSE HSG85 Electricity at work: Safe working practices

- HSE HSR25 Memorandum of guidance on the Electricity at Work Regulations 1989
- Electricity at Work Regulations 1989
- Electricity Safety, Quality and Continuity Regulations 2002

4 Definitions

Switching programme: a written, numbered sequence of switching operations giving the site, asset and expected result of each step, approved before use.

Operator: the Authorised Person (HV) who carries out the switching operations on site.

Repeat-back: the operator repeating an instruction word for word to the Control Engineer before acting on it.

Fault switching: switching instructed by the Control Engineer without a written programme, to restore supplies after a fault or to remove danger.

Normally open point: a switch kept open to separate two parts of the network that can be connected to transfer load.

Paralleling: closing a switch that connects two parts of the network supplied from different sources.

5 Responsibilities

Senior Authorised Person (HV): approves any programme on 33 kV plant, any programme involving GT-1 or GT-2, and any programme that parallels supplies from different primary substations.

Authorised Person (HV): writes programmes, independently checks programmes written by others, and acts as operator.

Control Engineer: approves programmes on 11 kV plant, issues each instruction, logs each operation and keeps the network diagram up to date.

Competent Person (HV): may assist the operator but does not carry out HV switching operations.

6 Procedure

6.1 Writing a switching programme

6.1.1 Write planned switching programmes on the standard GigaWhat Energy template at least five working days before the planned date.

6.1.2 Give each step a sequence number, the site ID, the asset ID and switch label, the operation, the expected state afterwards and the person who carries it out.

6.1.3 Include every isolation, safety lock, notice and Circuit Main Earth required by PR-ELEC-012, and the steps to remove them and restore normal running.

6.1.4 Check the loading on every circuit and transformer that will pick up load, using recent load data for the planned time of day.

6.1.5 Where one primary transformer, such as T-104 or T-105, will carry the whole substation load, confirm that the load is within its ONAF rating and that its cooling fans are in service.

6.1.6 Mark any step that parallels supplies and include the checks in 6.5.

6.1.7 Sign the programme as writer.

6.2 Checking and approving a programme

6.2.1 Have a second Authorised Person (HV) check the programme independently against the network diagram and site records, and sign it as checker.

6.2.2 Obtain approval from the Control Engineer for programmes on 11 kV plant.

6.2.3 Obtain approval from the Senior Authorised Person (HV) for any programme on 33 kV plant, involving GT-1 or GT-2, or paralleling supplies from different primary substations.

> HOLD POINT: A switching programme must not be used until it has been signed by the writer, the checker and the approver.

6.2.4 Do not amend an approved programme on site. If it cannot be followed, stop and contact the Control Engineer, who will either agree a written amendment with the original approver or cancel the programme.

6.3 Preparing to switch on site

6.3.1 Confirm the site and each item of equipment by reading the labels and checking them against the programme.

6.3.2 Check the switchgear for signs of distress before operating it, such as abnormal noise, a smell of burning, oil leaks, low gas pressure or a damaged mechanism.

6.3.3 Check the operational restrictions register for the switchboard, kept in the control room and at site.

6.3.4 Wear arc-rated clothing, a face visor and insulating gloves as required by PR-ELEC-001.

> WARNING: The 1974 oil circuit breakers on SWG-KGM-11 are under an operational restriction. A failure while opening or closing can cause an oil fire or explosion, and anyone in front of the panel could be killed.

6.3.5 Operate SWG-KGM-11 circuit breakers only from the control room, or with the portable remote operating device from outside the switchroom door, with nobody inside the switchroom.

6.3.6 Contact the Control Engineer, give the programme reference and confirm that you are ready to start.

6.4 Carrying out the switching

6.4.1 Receive one instruction at a time from the Control Engineer, quoting the programme step number.

6.4.2 Repeat back the instruction word for word and wait for the Control Engineer to confirm it is correct.

6.4.3 Carry out the operation, then check the position indicators and report the resulting state of the equipment.

6.4.4 Wait for the Control Engineer to log the time and confirm completion before receiving the next instruction.

> WARNING: A circuit breaker that fails to open or close correctly may be damaged internally and could fail violently if it is operated again.

6.4.5 If an operation gives an unexpected result, such as a circuit breaker failing to open, a protection trip or a wrong indication, stop, do not repeat the operation and report to the Control Engineer.

6.4.6 Resume only when the Control Engineer has agreed how to proceed, with a written amendment where the programme changes.

6.5 Load transfers and paralleling

6.5.1 Before closing a normally open point that parallels two sources, confirm with the Control Engineer that both sides are supplied from the same 33 kV system at ESS-RVL.

6.5.2 Check that the voltage difference across the open point is less than 3 percent, and do not close a point that has not been phased out since the last cable work on either side.

6.5.3 Keep the parallel for the shortest possible time and no longer than 15 minutes, then open the point named in the programme.

6.5.4 When transferring load between transformers, check the top-oil and winding temperatures of the transformer picking up load. If a temperature alarm appears, follow PR-ELEC-047.

6.5.5 Report the loading of the affected circuits and transformers to the Control Engineer after each transfer.

6.6 Fault switching

6.6.1 As Control Engineer, instruct fault switching without a written programme only to restore supplies under PR-ELEC-075, to take a transformer off load under PR-ELEC-047 or to remove danger.

6.6.2 Use the same instruction, repeat-back and confirmation as 6.4 for every fault switching operation.

6.6.3 As Control Engineer, record each fault switching operation in the control log at the time it is made.

6.6.4 Prepare a written programme under 6.1 and 6.2 before any isolation for a Permit-to-Work, even during a fault.

6.6.5 In an emergency to prevent danger to life, operate without instruction, then tell the Control Engineer immediately.

6.7 Completing the programme

6.7.1 Sign and time each step on the programme as it is completed.

6.7.2 Return the completed programme to the Control Engineer by the end of the shift.

6.7.3 As Control Engineer, update the network diagram and record any abnormal running arrangement left in place, with the reason.

6.7.4 Report any incident or near miss during switching under PR-CORP-001.

6.8 Emergency

6.8.1 If anyone is injured or there is a fire, call 999 and then inform the Control Engineer.

6.8.2 After a switchgear failure, leave the switchroom and do not re-enter until a Senior Authorised Person (HV) confirms it is safe.

6.8.3 Ask members of the public who report a power cut or damaged power lines to call 105.

7 Records

- Completed switching programmes, kept by the Control Engineer for six years.
- Control log entries for every planned and fault switching operation.
- Network diagram updates and the abnormal running arrangements log.

8 Revision history

Version	Date	Change
1	2010-06-07	First issue.
2	2013-06-03	Added independent checking by a second Authorised Person (HV).
3	2017-05-08	Added Senior Authorised Person (HV) approval for 33 kV programmes.
4	2021-06-07	Added the operational restrictions register check and remote operation of SWG-KGM-11 oil circuit breakers.
5	2025-06-02	Added paralleling checks and the 15-minute parallel limit. Added transformer temperature checks during load transfers and ONAF rating check for single-transformer running. Fault switching operations to be logged at the time they are made.